2013 Bored of Studies Trial Examinations

Mathematics Extension 2

General Instructions

- Reading time – 5 minutes.
- Working time – 3 hours.
- Write using black or blue pen. Black pen is preferred.
- Board-approved calculators may be used.
- A table of standard integrals is provided at the back of this paper.
- Show all necessary working in Questions 11 – 16.

Total Marks – 100

Section I Pages 1 – 5

10 marks

- Attempt Questions 1 – 10
- Allow about 15 minutes for this section.

Section II Pages 5 – 19

90 marks

- Attempt Questions 11 – 16
- Allow about 2 hours 45 minutes for this section.

Total marks – 10
Attempt Questions 1 – 10
All questions are of equal value

Shade your answers in the appropriate box in the Multiple Choice answer sheet provided.

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1 Let $P(x)$ be a cubic polynomial. Consider the following conditions.

(I) $\alpha^2+\beta^2+\gamma^2>0$.

(II) $\alpha^2+\beta^2+\gamma^2<0$.

(III) Has a root of multiplicity two.

Which of the following statements are always true?

(A) If (I) is true, then all three roots are real.

(B) If (II) is true, then there are at least two non-real roots.

(C) If (III) is true, then all three roots are real.

(D) None of the above.

2 Which of the following statements is always correct?

(A) If $z=a+ib$ is in the first quadrant, then $\arg(z)=\tan^{-1}\left(-\frac{b}{a}\right)$.

(B) If $z=a+ib$ is in the second quadrant, then $\arg(z)=\tan^{-1}\left(\frac{b}{a}\right)$.

(C) If $z=a+ib$ is in the third quadrant, then $\arg(z)=\tan^{-1}\left(\frac{b}{a}\right)$.

(D) If $z=a+ib$ is in the fourth quadrant, then $\arg(z)=\tan^{-1}\left(\frac{b}{a}\right)$.

3 A directrix of an ellipse has the equation $x = \frac{25}{4}$ and one of its foci has the coordinates $(-4, 0)$. What is the equation of this ellipse?

(A) $\frac{x^2}{5} + \frac{y^2}{3} = 1$.

(B) $\frac{x^2}{3} + \frac{y^2}{5} = 1$.

(C) $\frac{x^2}{25} + \frac{y^2}{9} = 1$.

(D) $\frac{x^2}{9} + \frac{y^2}{25} = 1$.

4 A convex polygon has 44 diagonals. How many sides does the polygon have?

(A) 8.

(B) 9.

(C) 10.

(D) 11.

5 Which of the following integrals is NOT equivalent to the others?

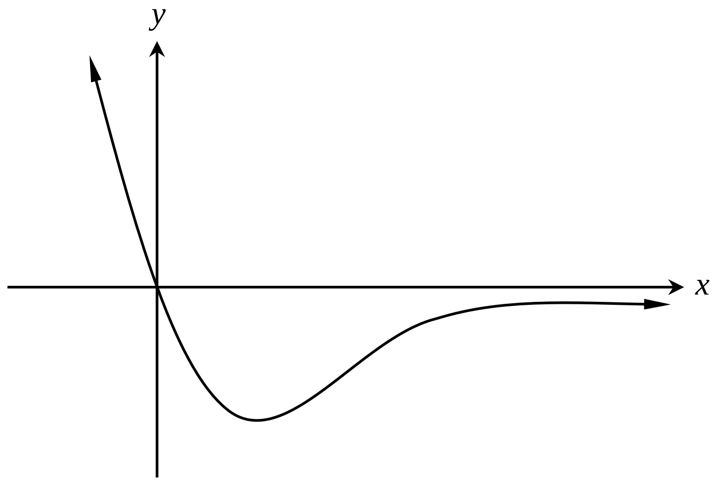
(A) $\int_0^{\frac{\pi}{4}} \tan^3 x \, dx$.

(B) $\int_0^1 \frac{x}{2(x+1)} dx$.

(C) $\int_1^e \frac{e^{2x}}{1+e^{-2x}} dx$.

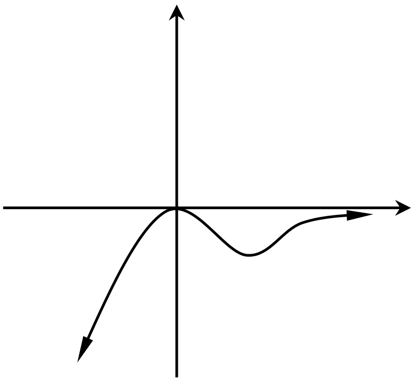
(D) $\int_0^1 \frac{x^3}{x^2+1} dx$.

6 The diagram shows a sketch of the curve $y = x\,f(x)$.

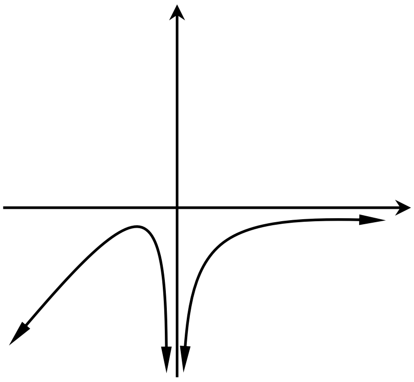


Which of the following best represents $y = f(x)$?

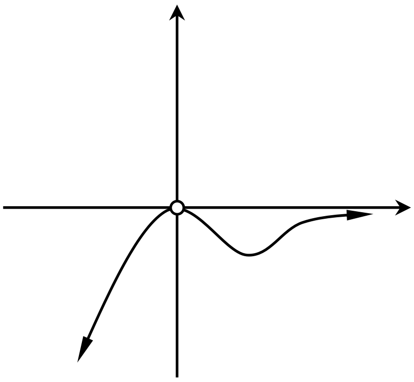
(A)



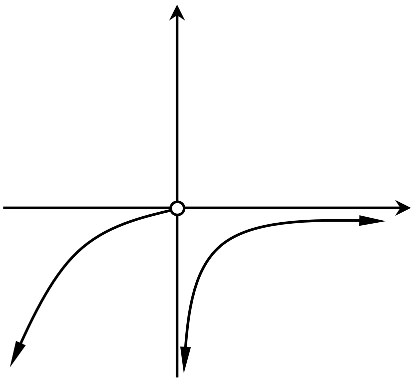
(C)



(B)



(D)



7 Which of the following polynomials is divisible by $x^2 - 2x + 1$?

(A) $P(x) = x^n - n(x-1) - 1$.

(B) $P(x) = x^n + n(x-1) - 1$.

(C) $P(x) = x^n - n(x-1) + 1$.

(D) $P(x) = x^n + n(x-1) + 1$.

8 A particle of mass m is undergoing circular motion in a circle of radius r and with angular velocity ω. Let the particle's tangential velocity be V and let its tangential acceleration be a. Which of the following expressions is correct?

(A) $a = mr\omega^2$.

(B) $a = r\omega^2$.

(C) $V = mr\omega$.

(D) $V = r\omega$.

9 The area bounded by the curve $y = \sin x$, the x axis and $x = \frac{\pi}{2}$ is rotated about the y axis. Which of the following is an expression for the volume of the solid formed?

(A) $\pi \int_0^{\frac{\pi}{2}} x^2 \cos x \, dx$.

(B) $2\pi \int_0^{\frac{\pi}{2}} x(1 - \sin x) \, dx$.

(C) $\pi \int_0^{\frac{\pi}{2}} x(\pi - x) \sin x \, dx$.

(D) $2\pi \int_0^{\frac{\pi}{2}} x \cos x \, dx$.

10 Suppose $f(x)$ is a continuous smooth function over $a \le x \le b$ and $g(x)$ is a continuous function over $c \le x \le d$. Which of the following integrals is always greater than or equal to the other choices?

(A) $\displaystyle\int_a^b f(x)\,dx + \int_c^d g(x)\,dx.$

(B) $\displaystyle\int_a^b |f(x)|\,dx + \int_c^d |g(x)|\,dx.$

(C) $\displaystyle\left|\int_a^b f(x)\,dx\right| + \left|\int_c^d g(x)\,dx\right|.$

(D) $\displaystyle\left|\int_a^b f(x)\,dx + \int_c^d g(x)\,dx\right|.$

Total marks – 90
Attempt Questions 11 – 16
All questions are of equal value

Answer each question in a SEPARATE writing booklet. Extra writing booklets are available.

Question 11 (15 marks) Use a SEPARATE writing booklet.

(a) Evaluate $\int_0^{\frac{\pi}{2}} \frac{a+b\sin x}{(b+a\sin x)^2}\,dx.$ **2**

(b) Use implicit differentiation to show that $x^2 - y^2 = a^2$ and $xy = c^2$, where $a,\ c \neq 0$, always intersect at right angles. **2**

(c) Let the origin, z_1, z_2 and z_3 be four points in the complex plane that form a cyclic quadrilateral with centre c.

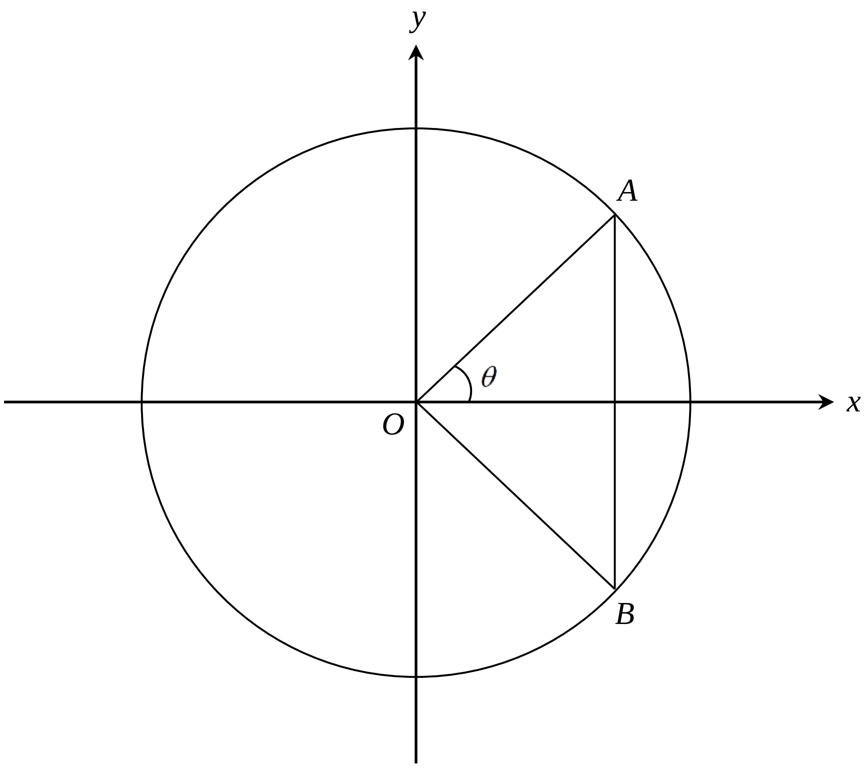
(i) Describe the set of complex numbers w satisfying $|w-k| = |w|$, where k is a fixed complex number. **1**

(ii) Prove that $\frac{1}{z_1}$, $\frac{1}{z_2}$ and $\frac{1}{z_3}$ are collinear. **2**

(d) Sketch the curve $y = \frac{2x^3-1}{2x^2-1}$. **3**

Question 11 continues on page 7

(e) In the circle $x^2 + y^2 = r^2$, an angle of $-\frac{\pi}{2} \leq 2\theta \leq \frac{\pi}{2}$ subtends sector *OAB*, which is symmetrical about the *x* axis. The areas described by ΔOAB and the minor segment *AB* are rotated about the *y* axis, forming solids with volumes V_1 and V_2 respectively.



(i) Find V_1 in terms of θ. **2**

(ii) Hence, find the value(s) of θ such that $V_1 = V_2$. **3**

End of Question 11

Question 12 (15 marks) Use a SEPARATE writing booklet.

(a) Define the sequence $a_{n+1} = 1 + a_1 a_2 \ldots a_n$ for all integers $n \geq 1$, where $a_1 = 1$.

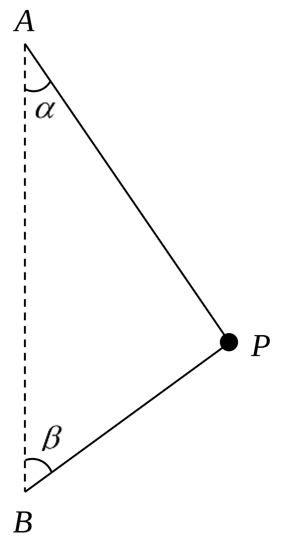
(i) Use Mathematical Induction to prove that **3**

$$\frac{1}{a_1} + \frac{1}{a_2} + \frac{1}{a_3} + \ldots + \frac{1}{a_n} = 2 - \frac{1}{a_1 a_2 a_3 \ldots a_n}.$$

(ii) Prove that $a_{n+1} > a_n$, for all $n \geq 1$. **1**

(iii) Hence, find the limiting value of $\sum_{k=1}^{n} \frac{1}{a_k}$ as $n \to \infty$. **1**

(b) A particle P of mass m spins with angular velocity ω in a circle of radius r, and is suspended by two light inextensible strings making angles from the vertical of α and β, where $0 < \alpha < \beta < \frac{\pi}{2}$. Let A be the point from which the top string is suspended from, and let B be the point where the bottom string is attached. **4**



The strings AP and BP experience tensions of T_1 and T_2 respectively.

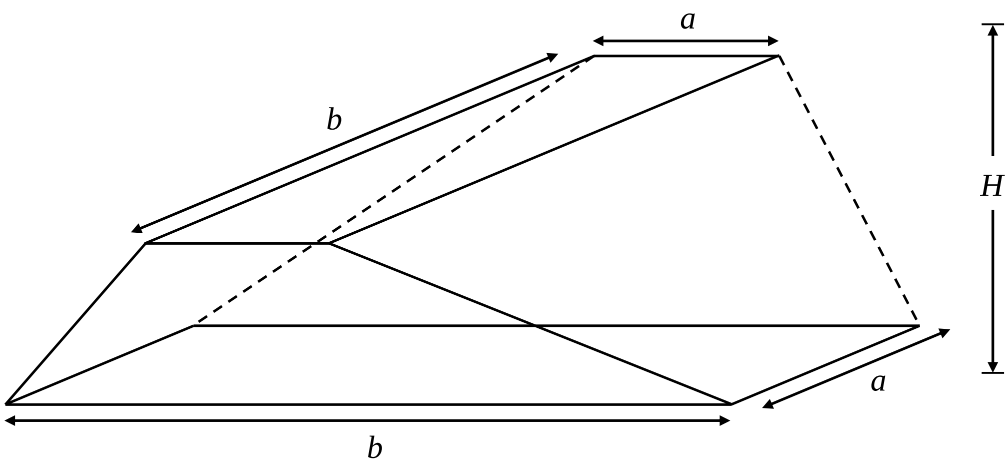
Prove that if $T_1 > T_2$, then $\omega^2 < \frac{g}{r}\left(\frac{\sin\alpha + \sin\beta}{\cos\alpha - \cos\beta}\right)$.

Question 12 continues on page 9

(c) Two sheets of paper with dimensions a and b, $a < b$, are stacked on a flat surface. One sheet of paper is raised up by a height H, and then rotated 90° about its centre.

The vertices of the bottom sheet and top sheet are connected, forming a solid.

Cross-sections taken parallel to the sheets are rectangles. Let the height of a typical cross section from the floor be h.



(i) Let δV be the volume of a cross section with thickness δh.

Show that **4**

$$\delta V = \frac{1}{H^2}\left[abH^2 + (b-a)^2\left(hH - h^2\right)\right]\delta h$$

(ii) Hence, show that the volume of the solid is **2**

$$V = \frac{H}{6}\left(a^2 + 4ab + b^2\right)$$

End of Question 12

Question 13 (15 marks) Use a SEPARATE writing booklet.

(a) A pack of n cards consists of q different colours. Each colour has p different cards labeled from 1 to p, such that $n = pq$.

From the pack, k cards are drawn out, where $2 < k \leq p < n$.

(i) Find the probability of exactly 2 of those cards being the same number. **2**

(ii) Show that if $p = q = k$, then the number of ways of having exactly two cards of the same number is $\frac{p^p}{2}(p-1)^2$. **1**

(b) A particle of mass m falls from rest and experiences a gravitational force of mg and a resistive force of mkv^2. Let its terminal velocity be denoted by u.

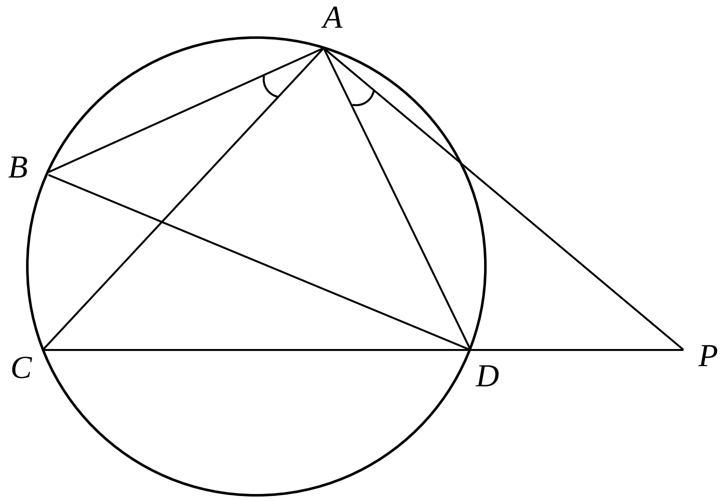
(i) Show that $u^2 = \frac{g}{k}$. **1**

(ii) Prove that the displacement of the particle at time t is **4**

$$x = \frac{1}{k}\ln\left(\frac{e^{\frac{gt}{u}} + e^{-\frac{gt}{u}}}{2}\right).$$

Question 13 continues on page 11

(c) Let $ABCD$ be a cyclic quadrilateral. A line is drawn through A to meet CD extended at P such that $\angle BAC = \angle DAP$.



(i) Prove that $\Delta BAC \parallel\!| \Delta DAP$ and hence show that $DP = \dfrac{AD \times BC}{AB}$. **2**

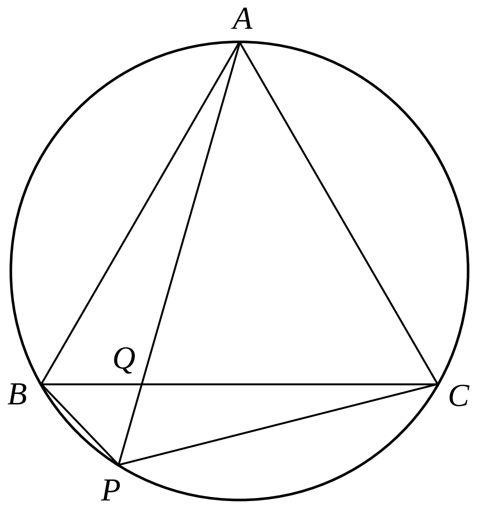
(ii) Prove that $\Delta ABD \parallel\!| \Delta ACP$ and hence show that $CP = \dfrac{AC \times BD}{AB}$. **2**

(iii) Prove that **1**

$$AC \times BD = AB \times CD + AD \times BC$$

Question 13 continues on page 12

Consider an equilateral triangle ABC inscribed within a circle. A line drawn from A passes through BC at Q and the circumference of the circle at P.



(iv) Use part (iii) to prove that $PA = PB + PC$. **2**

End of Question 13

Question 14 (15 marks) Use a SEPARATE writing booklet.

(a) Let $P(z) = z^{n+1} - kz^n + kz - 1$, where $-1 \le k \le 1$, and n is a positive integer.

(i) Prove that if α is a root of $P(z)$, $\dfrac{1}{\alpha}$ is also a root. **1**

(ii) Suppose there exists $n+1$ distinct roots of $P(z)$ all with modulus not equal to one. **1**

Explain why there exists at least one root α, such that $|\alpha| > 1$.

(iii) Show that $|\alpha|^{2n} |\alpha - k|^2 = |1 - k\alpha|^2$ **1**

(iv) Deduce that $\left(|\alpha|^2 - 1\right)\left(1 - k^2\right) < 0$. **2**

(v) Hence, explain why all roots of $P(z)$ have modulus 1. **1**

(b) Consider an n sided regular polygon centred at the origin with vertices $P_1P_2P_3...P_n$, where P_1 has coordinates $(1, 0)$.

Let $\theta = \dfrac{2\pi}{n}$ and define $d_k = P_1P_{k+1}$.

(i) Show that $d_k^2 = (1 - \cos k\theta)^2 + \sin^2 k\theta$ **1**

(ii) Hence, show that **1**

$$d_k^2 = 2 - \omega^k - \omega^{n-k},$$

where ω is an n^{th} root of unity.

(iii) Deduce that $d_1^2 + d_2^2 + d_3^2 + ... + d_{n-1}^2 = 2n$. **2**

Question 14 continues on page 14

(c) Let a, b, A and B be positive numbers.

(i) Prove that $\dfrac{ab}{AB} \le \dfrac{1}{2}\left(\dfrac{a^2}{A^2} + \dfrac{b^2}{B^2}\right)$. **1**

(ii) Let $A = \sqrt{\sum_{k=1}^{n} a_k^2}$ and $B = \sqrt{\sum_{k=1}^{n} b_k^2}$, where a_k and b_k are positive real numbers. **2**

Use (i) to prove that

$$\left(\sum_{k=1}^{n} a_k b_k\right)^2 \le \left(\sum_{k=1}^{n} a_k^2\right)\left(\sum_{k=1}^{n} b_k^2\right)$$

(iii) Let $S = x_1 + x_2 + x_3 + \ldots + x_n$, where $x_k > 0$ for all $1 \le k \le n$. **2**

Use (ii) to prove that

$$\frac{S}{S - x_1} + \frac{S}{S - x_2} + \frac{S}{S - x_3} + \ldots + \frac{S}{S - x_n} \ge \frac{n^2}{n-1}$$

End of Question 14

Question 15 (15 marks) Use a SEPARATE writing booklet.

(a) Define the series $S_n(x) = \frac{1}{2} + \sum_{k=1}^{n} \cos(kx)$, where $0 \le x \le \pi$.

(i) Prove that $\sin(\alpha+\beta) - \sin(\alpha-\beta) = 2\cos\alpha\sin\beta$ **1**

(ii) Prove that **2**

$$S_n(x) = \begin{cases} \dfrac{\sin\left[(2n+1)x/2\right]}{2\sin(x/2)}, & \text{if } x \ne 0. \\ n + \dfrac{1}{2}, & \text{if } x = 0. \end{cases}$$

(b) Prove that $\int_0^{\pi} x\cos(kx)\,dx = \frac{1}{k^2}\left[(-1)^k - 1\right]$, where k is a positive integer. **2**

(c) Define the integral $I_n = \lim_{a \to 0} \int_a^{\pi} x\, f_n(x)\,dx$, where

$$f_n(x) = \frac{\sin\left[(2n+1)x/2\right]}{2\sin(x/2)}$$

(i) Use (a) and (b), or otherwise, to prove that **1**

$$I_n = \frac{\pi^2}{4} + \sum_{k=1}^{n}\left[\frac{(-1)^k - 1}{k^2}\right].$$

(ii) Deduce that **1**

$$\frac{1}{2} I_{2n-1} = \frac{\pi^2}{8} - \sum_{k=1}^{n} \frac{1}{(2k-1)^2}.$$

Question 15 continues on page 16

Let $g(x)=\dfrac{d}{dx}\left(\dfrac{x/2}{\sin(x/2)}\right)$.

(iii) Using Integration by Parts, or otherwise, prove that **3**

$$I_{2n-1}=\frac{2}{4n-1}\left(1+\lim_{a\to 0}\int_a^{\pi} g(x)\cos\left[\frac{(4n-1)x}{2}\right]dx\right).$$

(iv) Prove that **2**

$$1-\frac{\pi}{2}\le \lim_{a\to 0}\int_a^{\pi} g(x)\cos\left[\frac{(4n-1)x}{2}\right]dx\le\frac{\pi}{2}-1,$$

(v) Deduce that as $n\to\infty$, $I_{2n-1}\to 0$. **1**

(vi) Hence, prove that **2**

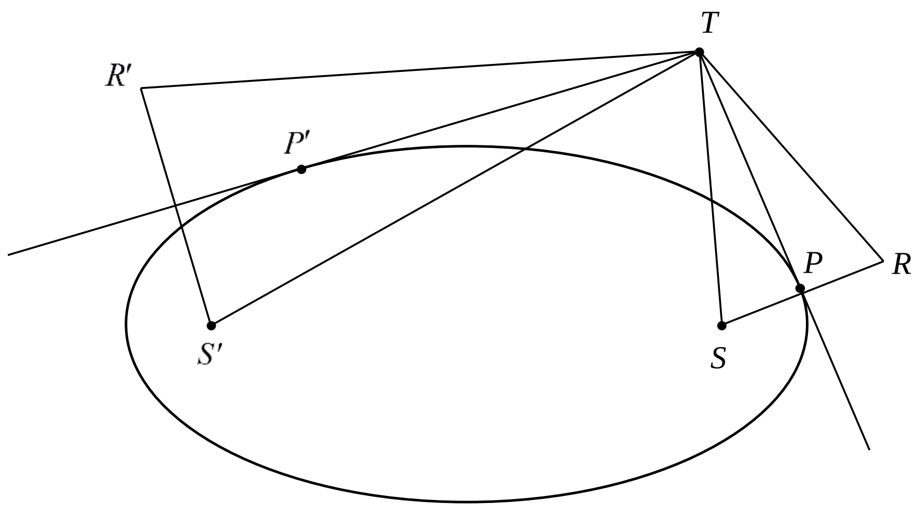
$$\sum_{k=1}^{\infty}\frac{1}{k^2}=\frac{\pi^2}{6}.$$

End of Question 15

Question 16 (15 marks) Use a SEPARATE writing booklet.

(a) Let T be an external point of an ellipse. From T, two tangents are drawn to the ellipse and let P and P' be the points of contact. Lines are drawn from T to both foci S and S'.

S and S' are reflected about the lines TP and TP' respectively, forming R and R' respectively.



(i) Prove that S, P' and R' are collinear. **2**

(ii) Hence, prove that $\Delta TR'S \equiv \Delta TS'R$. **2**

(iii) Deduce that $\angle STP = \angle S'TP'$. **1**

Question 16 continues on page 18

(b) Let $P(z)$ be a polynomial with real coefficients with exactly one real root. Its roots z_1, z_2 and z_3 form the vertices of a triangle Δ_P in the complex plane.

Define $T(z) = Az + B$, where A is a real non-zero constant, and B is a fixed complex number.

Let Δ_T be the triangle with vertices $T(z_k)$, where $k = 1, 2, 3$.

(i) Explain why $\Delta_T \,|||\, \Delta_P$. **1**

(ii) Let $P_T(z) = (z - T(z_1))(z - T(z_2))(z - T(z_3))$. **1**

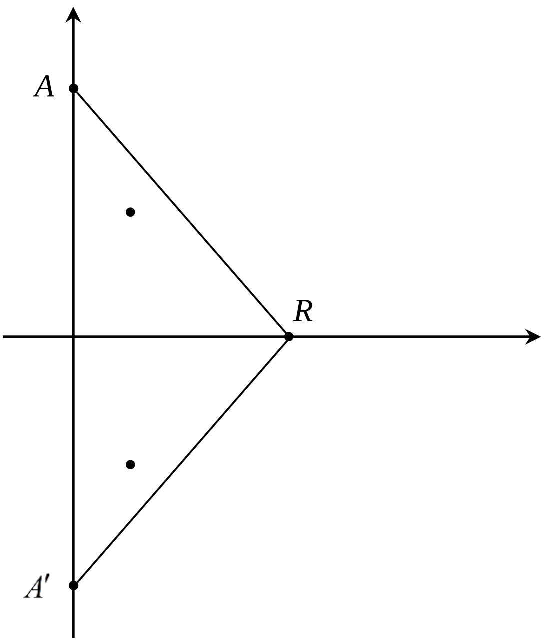
Prove that $P_T(T(z)) = A^3 P(z)$.

(iii) Deduce that if α is a root of $P'(z)$, then $T(\alpha)$ is a root of $P_T'(z)$. **1**

(iv) Describe the geometric significance of (iii). **1**

Question 16 continues on page 19

(c) Let $P(z)$ be a monic cubic polynomial with roots $i, -i, r$, where r is a positive non-zero real number. Let the triangle formed by the roots be T with vertices A, A', R.



(i) Show that $P'(z) = 3z^2 - 2rz + 1$. **1**

(ii) Suppose $P'(z)$ has no real roots. **3**

Prove that the roots of $P'(z)$ lie within T.

(d) Let $S(x)$ be a real cubic polynomial with two non-real roots such that the roots of $S'(x)$ are also non-real. Let the roots of $S(x)$ be z_1, z_2, z_3 and let the roots of $S'(x)$ be w_1 and w_2. **2**

Use (b) and (c) to show that w_1 and w_2 always lie within the triangle defined by z_1, z_2, z_3.

End of Exam

STANDARD INTEGRALS

$$\int x^n\,dx = \frac{1}{n+1}x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x}\,dx = \ln x, \quad x > 0$$

$$\int e^{ax}\,dx = \frac{1}{a}e^{ax}, \quad a \neq 0$$

$$\int \cos ax\,dx = \frac{1}{a}\sin ax, \quad a \neq 0$$

$$\int \sin ax\,dx = -\frac{1}{a}\cos ax, \quad a \neq 0$$

$$\int \sec^2 ax\,dx = \frac{1}{a}\tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax\,dx = \frac{1}{a}\sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2+x^2}\,dx = \frac{1}{a}\tan^{-1}\frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2-x^2}}\,dx = \sin^{-1}\frac{x}{a}, \quad a \neq 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2-a^2}}\,dx = \ln\left(x+\sqrt{x^2-a^2}\right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2+a^2}}\,dx = \ln\left(x+\sqrt{x^2+a^2}\right)$$

NOTE: $\ln x = \log_e x, \quad x > 0$